

Sentiment Analysis of X (Twitter) Data

Introduction

Social media platforms have become one of the most important sources for expressing opinions, thoughts, and emotions. Millions of users share their views every day on different topics, making social media data a valuable resource for understanding public sentiment. Sentiment analysis is a technique used to identify and classify opinions expressed in text as positive, negative, or neutral.

In this project, sentiment analysis was performed on a dataset containing tweets collected from X (formerly Twitter). The main objective was to analyze the sentiment of tweets and understand the overall distribution of positive, negative, and neutral opinions. Using Python and its data analysis libraries, the dataset was explored, visualized, and interpreted to gain meaningful insights into user sentiments.

This project demonstrates how sentiment analysis can be used to better understand public opinion and support data-driven decision-making through social media analytics.

Dataset Overview

The dataset used in this project contains tweets collected from X (formerly Twitter) along with their sentiment labels. The data consists of two main columns: `clean_text`, which contains the processed tweet text, and `category`, which represents the sentiment associated with each tweet.

The sentiment labels are categorized as:

- 1 = Positive
- 0 = Neutral
- -1 = Negative

The dataset was analyzed using Python libraries such as Pandas, Matplotlib, and Seaborn. Before performing sentiment analysis, the dataset was explored to understand its structure and sentiment distribution.

Dataset Information

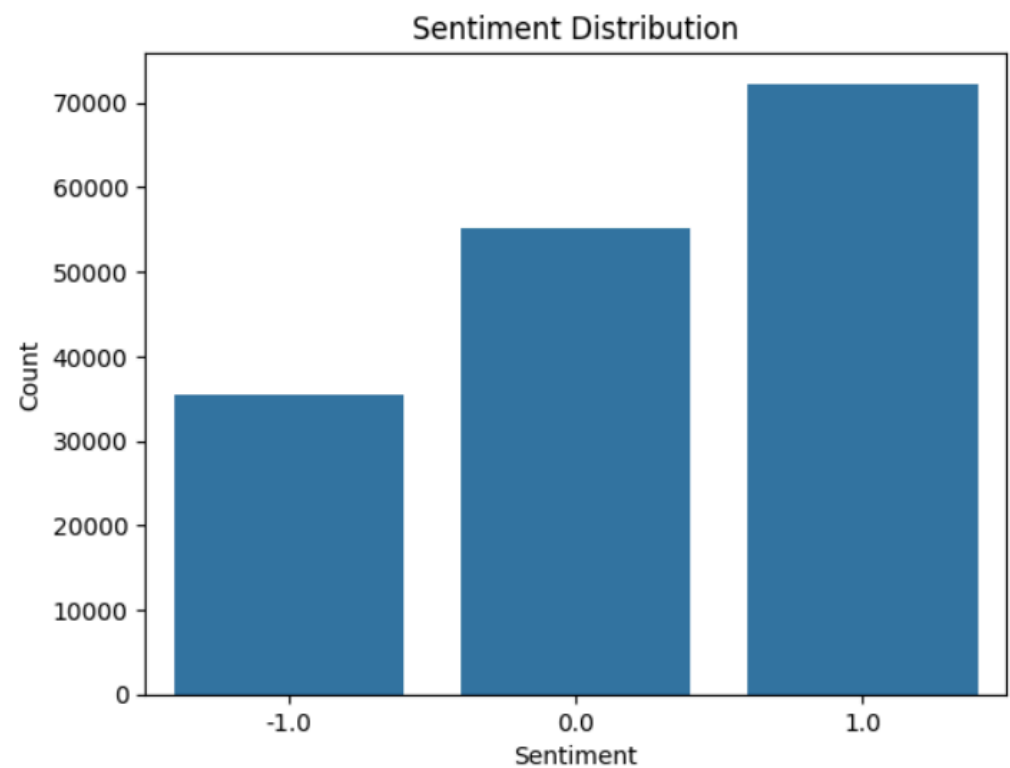
The dataset contains 162,980 records and 2 columns. The `clean_text` column stores tweet content, while the `category` column stores sentiment labels. The dataset includes positive, neutral, and negative tweets, providing a balanced source for sentiment analysis.

	clean_text	category
0	when modi promised "minimum government maximum...	-1.0
1	talk all the nonsense and continue all the dra...	0.0
2	what did just say vote for modi welcome bjp t...	1.0
3	asking his supporters prefix chowkidar their n...	1.0
4	answer who among these the most powerful world...	1.0

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 162980 entries, 0 to 162979
Data columns (total 2 columns):
#   Column      Non-Null Count  Dtype
---  -
0   clean_text  162976 non-null  object
1   category    162973 non-null  float64
dtypes: float64(1), object(1)
memory usage: 2.5+ MB
```

Sentiment Distribution Analysis

To understand the overall sentiment present in the dataset, the frequency of each sentiment category was calculated. The analysis showed that positive tweets were the most common, followed by neutral tweets and negative tweets. This distribution provides an overview of how users express their opinions on the platform.

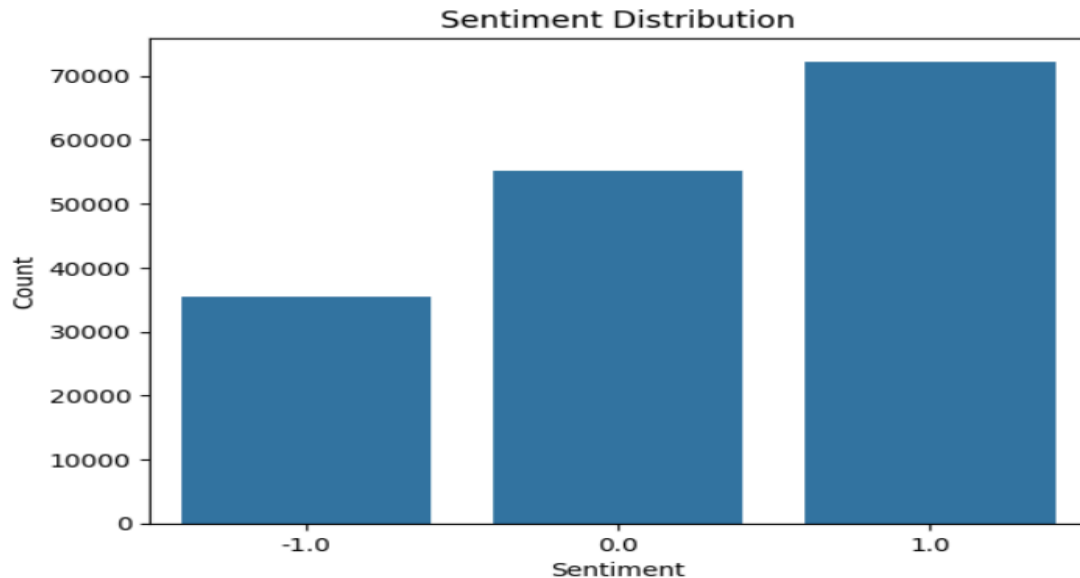


```
df['category'].value_counts()
```

	clean_text	category	Sentiment
0	when modi promised "minimum government maximum...	-1.0	Negative
1	talk all the nonsense and continue all the dra...	0.0	Neutral
2	what did just say vote for modi welcome bjp t...	1.0	Positive
3	asking his supporters prefix chowkidar their n...	1.0	Positive
4	answer who among these the most powerful world...	1.0	Positive

Visualization 1: Sentiment Distribution

A count plot was created to visualize the distribution of sentiment categories in the dataset. The graph clearly shows the number of positive, neutral, and negative tweets.



Visualization 2: Positive, Neutral and Negative Tweets

The sentiment labels were converted into meaningful categories to improve readability. A second count plot was created to provide a clearer understanding of sentiment distribution among the tweets.

Findings

- Positive tweets were the highest in number.
- Neutral tweets formed a significant portion of the dataset.
- Negative tweets were comparatively lower.
- Social media sentiment analysis helps understand public opinion and user behavior.
- Visualization techniques made it easier to interpret sentiment patterns.

Conclusion

The sentiment analysis of X (Twitter) data provided valuable insights into the opinions and emotions expressed by users on social media. By analyzing the sentiment labels and visualizing the data, it was possible to understand the overall distribution of positive, neutral, and negative tweets. The results showed that positive sentiments were more common compared to negative sentiments, indicating a generally favorable tone within the dataset.

Using Python libraries such as Pandas, Matplotlib, and Seaborn made it easier to explore, analyze, and visualize the data effectively. This project demonstrated the importance of sentiment analysis in understanding public opinion and highlighted how social media data can be used to gain meaningful insights for research, business, and decision-making purposes.